DSA TASKS — TYPED NOTES

Tasks 1, 2 and 3

TASK 1 — Identify Variables and Their Data Types

Question: Identify the list of variables and their data types for the following:

1. Bill of an Amazon Order

• Seller GST Code → Integer

• Address → Dictionary

• Place of Supply → String

• Place of Delivery → String

• Invoice Number → String

• Invoice Date → String

• PAN No. / Registration No. → String

• Description → Object

• Total → Float

• Amount in Words → String

2. Bus Ticket

• Depot → String

• Date → String

• Price → Float

• PNR No. → Integer

• KMS → Integer

• E-POS → Integer

• Vehicle No. → String

3. Current Bill

• Date → String

• Time → String

• ERO/SEC → String

• BNO → Integer

• Area Code → Integer

• Name → String

• Recorded M.D. → Float

• MF/PF/Billed M.D. → Float

• Bill & All Charges → Float

4. Gas Bill

- Consumer No. → Integer
- Distributor Serial No. → String
- Name → String
- Address → Dictionary
- Order No. → String
- Order Date → String
- Connection Type → String
- Subsidy Confirmed → String
- Price → Float
- Order Status/Type → String

TASK 2 — DSA, Real-World Data Structures and C Structure Size

1. Companies conducting placement drives using DSA

Top Product Companies: Google, Microsoft, Amazon, Apple, Meta, Adobe, Oracle, Salesforce, Atlassian, ServiceNow, Uber, Nvidia.

India Product Companies: Flipkart, PhonePe, Swiggy, Zepto, Meesho, Razorpay, Paytm, Zoho, Freshworks.

Service-Based Companies: Tata Consultancy Services (TCS), Infosys, Wipro, HCLTech, Tech Mahindra, Accenture, Cognizant, Capgemini, LTIMindtree, IBM.

2. GATE / DSA syllabus

• C programming basics

• Time complexity

• Arrays & strings

• Linked lists

• Stacks & queues

• Trees & BST

• Heaps

• Graphs

• Searching & sorting

• Hashing

• Recursion

• Divide and conquer

• Greedy algorithms

• Dynamic programming

• Graph algorithms (BFS, DFS, MST, shortest path)

• Pointers

• Structures

• Self-referential structures

• Functions

• Storage classes

3. Inputs and Outputs

Electricity Bill — Inputs: Consumer ID, Consumer Name, Meter Number, Previous Reading, Current Reading, Cost per Unit.
Outputs: Consumer ID, Consumer Name, Units Consumed, Total Bill Amount.

Bus Ticket — Inputs: Source, Destination, Bus/Route Number, Ticket Fare, Payment Amount.
Outputs: Ticket Number, Source, Destination, Bus Number, Fare, Date & Time, Balance Amount, Ticket Printed.

WhatsApp Message — Inputs: Friend's phone number, Message Text, Emojis, Image/Video/Document, Send button.

Outputs: Message Sent, Message Delivered, Message Read, Sent Time.

4. Data Structures

• Students standing in mess queue → Queue

• Contact list in phone → Array / List

• Roads connecting hostel and college → Graph

• Books in a bag → Stack

• College departments → Tree

• Hostel room numbers → Array

• Biometric attendance in mess → Hash Table

• Hostel attendance register → Array / List

5. C Structure Variable and sizeof()

Create a structure variable **student** and estimate the size of the structure manually, then write a C program to find the size using sizeof operator.

```
#include
struct student {
    int rollno;
    char grade;
    float marks;
};
int main()
{
    struct student s;
    printf("size of student structure = %d", sizeof(s));
    return 0;
}
```

Manual size: int = 4 bytes; char = 1 byte; float = 4 bytes. Total = 9 bytes.

sizeof() result: int = 4 bytes; char = 1 byte; padding = 3 bytes; float = 4 bytes. Total = 12 bytes.

Conclusion: The compiler inserts padding for memory alignment, so the manual size and sizeof() result are different.

TASK 3 — Amazon Functions and return Values in C

1. Ten different functions on Amazon's website

- User Registration
- User Login
- Search Products
- Browse Categories
- Add to Cart
- Buy Now
- Wishlist
- Online Payment
- Order Tracking
- Product Reviews & Ratings

2. return 1 and return 0

return 0; — Indicates that the program executed successfully. main() returns control to the operating system.

return 1; — Indicates that the program ended with an error or unsuccessfully. main() also returns control to the operating system.

The return value tells the operating system whether the program finished successfully or with an error.

ADDITIONAL TASK — C STRUCTURE AND sizeof()

Question

Create a structure variable **student** and estimate the size of the structure manually. Then write a C program to find the size of a structure using the **sizeof** operator. Observe both outputs. Are they the same or different? Justify your answer.

Structure members

int rollno;

char grade;

float marks;

The task compares the manually calculated size with the size returned by the C program.

C PROGRAM AND SIZE CALCULATION

C Program

```
#include
struct student
{
    int rollno;
    char grade;
    float marks;
};
int main()
{
    struct student s;
    printf("Size of structure = %d bytes\n", sizeof(s));
    return 0;
}
```

Manual Calculation

int = 4 bytes
char = 1 byte
float = 4 bytes
Total = 9 bytes

Program Output

Size of structure = 12 bytes

Why are the outputs different?

The compiler inserts extra unused bytes between members or at the end of the structure for proper memory alignment. Therefore, the manually calculated size and sizeof() result can be different.

TASK — JAVA HASH CODE

Question

In Java every object has a hash code. Find out what is hash code and why it is being generated in Java.

What is hash code?

A hash code is an integer value that represents an object. It is mainly used to identify objects efficiently in hash-based collections such as **HashMap, HashSet and Hashtable**.

Why is it generated?

• Improve searching speed.

• Help compare objects efficiently.

• Determine the bucket in a hash-based collection.

Example

```
class Demo
{
    public static void main(String[] args)
    {
        String s = "Hello";
        System.out.println(s.hashCode());
    }
```

Output: 69609650